

Hyper-Differential Operator-Based Graph Fractional Fourier Transform: Novel Approaches and Applications

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Abstract—This paper introduces novel approaches and applications for hyper-differential operator-based graph fractional Fourier transform (HGFrFT), which provides a more flexible and efficient framework for graph signal processing. Its mathematical foundation is derived through the relationship between the graph Fourier transform and the fractional graph Fourier transform, thereby enhancing its ability to represent the underlying structure of graph signals. Specifically, four solutions to the Sylvester equation are discussed, and two fixed hyperdifferential operator matrices are introduced to define the HGFrFT. Furthermore, the applications of the HGFrFT in vertex-frequency analysis and optimal filter design are explored. Finally, the effectiveness of the HGFrFT is validated in electrocardiogram signal classification, demonstrating its applicability to real-world signal processing tasks. The theoretical framework and experimental results collectively advance fractional operator theory in graph domains.

Index Terms—Graph signal processing, graph Fourier transform, graph fractional Fourier transform, operator theory, fractional Fourier transform.

I. INTRODUCTION

WITH the rapid development of information technology, a vast amount of data characterized by irregularity and discreteness has emerged in various fields, including social networks, biological systems, and sensor networks. Traditional signal processing techniques, primarily designed for structured data, encounter substantial difficulties when dealing with unstructured data [1]. Graph signal processing (GSP) has emerged as a robust framework for managing irregular data structures by utilizing the mathematical principles of graph theory and spectral analysis [2], [3]. GSP extends classical signal processing concepts to graph-defined signals, facilitating their efficient

representation, filtering, and transformation within complex networked systems [4], [5], [6]. Given the increasing prevalence of graph-structured data in modern applications, advancing GSP techniques is essential for overcoming the limitations of traditional approaches.

The core of GSP is rooted in the graph Fourier transform (GFT), which is formulated through spectral decomposition of graph shift operators, such as the adjacency matrix or Laplacian matrix, to derive the transformation matrix (graph Fourier basis) [6], [7]. Different graph Fourier bases correspond to varying degrees of smoothness, with their associated eigenvalues characterizing the gradient of graph signal smoothness. Consequently, these eigenvalues are interpreted as frequencies. The coefficients of a graph signal in different graph Fourier bases correspond to those of the GFT, quantifying the magnitude of signal components at respective frequencies and indicating the signal's intensity at these frequencies [1], [5], [8]. However, for complex signal structures, the fixed smoothing patterns of GFT, which rely solely on the shift operator, are limited in fully capturing the intrinsic geometric properties of the signal, thus encountering performance bottlenecks in improving signal processing efficiency [9].

As a natural extension of classical Fourier analysis, the fractional Fourier transform (FrFT) introduces a fractional domain that interpolates between the time and frequency domains, providing greater flexibility and broader applicability [10], [11], [12], [13]. By enabling the analysis of signals within an intermediate domain between time and frequency, FrFT establishes a more generalized transformation framework, which proves particularly beneficial for non-stationary signal processing, optical information processing, and time-varying system analysis [14], [15], [16], [17], [18], [19], [20], [21]. Its capability to adjustably balance time and frequency resolution renders it a versatile tool in applications such as radar signal processing, image enhancement, and machine learning [22], [23], [24].

Many researchers have extended fractional domain transforms to GSP, facilitating the selection of different smoothing modes for fractional graph Fourier bases [9], [25], [26], [27]. This flexibility enables an optimal alignment with the signal's intrinsic local features, offering novel perspectives and methodologies for problem-solving and fostering numerous innovative applications. For instance, the GFrFT has been investigated in vertex-frequency analysis, filter design, high-dimensional analysis, and various practical applications [28], [29], [30].

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These advancements significantly enhance the capabilities of graph-based data processing, allowing for finer-grained control over signal representation. Overall, GFrFT broadens the scope of traditional GSP by incorporating fractional domain adaptability, laying the groundwork for enhanced analytical techniques and expanding its applications in science and engineering [31], [32].

Traditional GFrFT extends the conventional FrFT to graph signals through the incorporation of a fractional parameter into the GFT matrix [9]. However, this approach is dependent on matrix eigendecomposition, which results in high computational complexity and potential numerical instability. To mitigate these limitations, Tuna Alikasıfoğlu et al. proposed a GFrFT based on hyperdifferential operators [33], [34], [35], [36], which reformulates the transformation into an exponential operator form, providing both computational efficiency and a more intuitive mathematical framework [37]. Building upon the foundational work in [37], the hyperdifferential operator-based GFrFT (denoted as HGFrFT) is systematically derived by establishing its connection to both traditional GFrFT and GFT. In particular, this work contributes four distinct solution strategies to the Sylvester equation for defining the hyperdifferential operators, each offering different trade-offs in terms of accuracy and computational cost. To further enhance the applicability and efficiency of the framework, we introduce two fixed operator matrix forms, providing simplified, efficient approximations suitable for real-time or large-scale applications. Additionally, we perform an extensive analysis and comparison of these approaches through simulations, covering important aspects such as numerical stability and performance under noise. Furthermore, associated vertex-frequency analysis techniques and optimal filter designs are developed. Finally, the effectiveness of the HGFrFT in electrocardiogram (ECG) signal classification is validated, highlighting its potential in real-world applications.

This paper is organized as follows. Section II introduces the fundamental concepts of GSP and the FrFT. Section III establishes the theoretical foundation of the HGFrFT by linking GFrFT with GFT and deriving its formal definition using different approaches to solving the Sylvester equation. Section IV presents numerical experiments and analysis on equivalence with the traditional GFrFT, computational efficiency, numerical stability, and graph signal examples. Section V investigates parameter optimization for the fractional order. Section VI explores the HGFrFT-based vertex-frequency analysis and optimal filter design. Section VII examines its application in ECG signal classification. Finally, Section VIII presents the conclusions of this study.

II. PRELIMINARIES

A. Graph Fourier Transform

A graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{A}\}$ is composed of a vertex set $\mathcal{V} = \{v_0, v_1, \dots, v_{N-1}\}$ with N elements, an edge set $\mathcal{E}$, and an adjacency matrix $\mathbf{A}$ [7]. The graph Laplacian $\mathcal{L}$ can be expressed as $\mathcal{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ denotes the diagonal degree matrix. Each diagonal entry d_i of $\mathbf{D}$ represents the total sum of the weights of edges connected to vertex i . A graph signal $\mathbf{f} \in \mathbb{R}^N$

is defined as a mapping from the vertex set $\mathcal{V}$ to real values, with each vertex v_i assigned a value $f(i)$.

The adjacency matrix $\mathbf{A}$, the Laplacian $\mathcal{L}$, the row-normalized adjacency matrix $\mathbf{A}_{\text{row}} = \mathbf{D}^{-1}\mathbf{A}$, and the symmetric normalized Laplacian $\mathcal{L}_{\text{norm}} = \mathbf{I}_N - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ are examples of graph shift operators, collectively denoted by $\mathcal{K}$. The operator $\mathcal{K}$ is given by $\mathcal{K} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$, where $\mathbf{\Lambda} = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$ represents its eigenvalues. The eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The columns of the matrix $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}]$ are composed of the generalized eigenvectors of $\mathcal{K}$. The GFT of a signal $\mathbf{f} \in \mathbb{R}^N$ defined on the vertices of the graph $\mathcal{G}$ is given by [6]

$$\hat{\mathbf{f}} = \mathcal{F}\mathbf{f} = \mathbf{U}^{-1}\mathbf{f}. \quad (1)$$

The inverse GFT is given by

$$\mathbf{f} = \mathcal{F}^{-1}\hat{\mathbf{f}} = \mathbf{U}\hat{\mathbf{f}}. \quad (2)$$

B. Fractional Fourier Transform

The FrFT of a signal $f(t) \in L^2(\mathbb{R})$ is defined as [10], [11]

$$\mathcal{F}_{c\alpha}[f](u) = \int_{-\infty}^{\infty} f(t)K_{\alpha}(u, t) dt, \quad (3)$$

where the kernel function $K_{\alpha}(t, u)$ is given by

$$K_{\alpha}(u, t) = \begin{cases} A_{\alpha} \exp \left\{ i \frac{u^2}{2} \cot(\alpha) - itu \csc(\alpha) \right. \\ \quad \left. + i \frac{t^2}{2} \cot(\alpha) \right\}, \alpha \neq n\pi, \\ \delta(u - t), \alpha = 2n\pi, \\ \delta(u + t), \alpha = 2n\pi - \pi. \end{cases} \quad (4)$$

The coefficient A_{α} is defined as $A_{\alpha} = \sqrt{\frac{1-i\cot(\alpha)}{2\pi}}$, $\alpha = \frac{p\pi}{2}$, $0 < |p| < 2$, $0 < |\alpha| < \pi$, where i represents the imaginary unit.

A hyper-differential operator is defined by the inclusion of differential operators within an exponential function. The classical FrFT operator $\mathcal{F}_{c\alpha}$ can be represented in hyper-differential form as [34], [38]

$$\mathcal{F}_{c\alpha} = \exp \left(-i\alpha\pi^2 \frac{\mathcal{P}_c^2 + \mathcal{Q}_c^2}{2} \right), \quad (5)$$

where $\mathcal{Q}_c f(u) = uf(u)$ is the coordinate multiplication operator, and $\mathcal{P}_c f(u) = \frac{1}{2\pi i} \frac{df(u)}{du}$ is the differential operator. These two operators satisfy a Fourier duality, as expressed by [35]

$$\mathcal{P}_c = \mathcal{F}_c \mathcal{Q}_c \mathcal{F}_c^{-1}, \quad \mathcal{Q}_c = \mathcal{F}_c \mathcal{P}_c \mathcal{F}_c^{-1}, \quad (6)$$

where $\mathcal{F}_c$ denotes the classical Fourier transform operator. The discrete counterpart of the FrFT, formulated using hyper-differential operators, is given by [33], [36]

$$\mathcal{F}_{\alpha} = \exp \left(-i\alpha\pi^2 \frac{\mathcal{P}^2 + \mathcal{Q}^2}{2} \right), \quad (7)$$

where $\mathcal{Q}$ and $\mathcal{P}$ represent the discrete analogs of the coordinate multiplication and differentiation operators.

III. GRAPH FRACTIONAL FOURIER TRANSFORM

The GFrFT has been recognized as an essential and versatile tool in GSP [9], [31]. By facilitating the introduction of an intermediate domain between the graph vertex and spectral domains, the GFrFT offers novel perspectives for interpreting and manipulating graph signals. This intermediate representation is instrumental in applications that require multi-resolution analysis or involve signal characteristics that gradually vary across the graph [28], [29]. In recent years, this innovative approach has garnered significant attention, thereby fostering numerous advancements in both theoretical frameworks and practical implementations [26], [37].

Let $\mathcal{F} = \mathbf{U}^{-1} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^{-1}$, where $\mathbf{\Sigma} = \text{diag}(\sigma_0, \sigma_1, \dots, \sigma_{N-1})$ is the diagonal matrix of eigenvalues. Here, $\mathbf{U}$ denotes the eigenvector matrix of the graph shift operator $\mathcal{K}$, while $\mathbf{V}$ represents the eigenvector matrix of $\mathcal{F}$ itself. The GFrFT is defined as [9]

$$\hat{\mathbf{f}}_\alpha = \mathcal{F}_\alpha \mathbf{f} = (\mathbf{U}^{-1})^\alpha \mathbf{f} = \mathbf{V}\mathbf{\Sigma}^\alpha \mathbf{V}^{-1} \mathbf{f}, \quad (8)$$

where $\mathcal{F}_\alpha = (\mathbf{U}^{-1})^\alpha = \mathbf{V}\mathbf{\Sigma}^\alpha \mathbf{V}^{-1}$, with α denoting the fractional order. The inverse GFrFT is given by

$$\mathbf{f} = \mathcal{F}_\alpha^{-1} \hat{\mathbf{f}}_\alpha = \mathbf{V}\mathbf{\Sigma}^{-\alpha} \mathbf{V}^{-1} \hat{\mathbf{f}}_\alpha. \quad (9)$$

The above definition utilizes the characteristic function and eigenvalue properties of the FrFT, extending these principles to GSP.

Analogous to the hyper-differential operator framework in the conventional FrFT [33], [34], [35], [36], constructing a corresponding hyper-differential operator representation for the GFrFT in GSP remains a significant and challenging area of ongoing research [37]. In traditional signal processing, the fractional Fourier operator can be expressed as $\mathcal{F}_{c\alpha} = \exp(-i\alpha\pi^2(\mathcal{P}_c^2 + \mathcal{Q}_c^2)/2)$, where $\mathcal{P}_c$ and $\mathcal{Q}_c$ are differential and coordinate multiplication operators satisfying Fourier duality.

Motivated by this formulation, we seek to construct graph signal operators $\mathcal{P}$ and $\mathcal{Q}$ that satisfy analogous duality relationships through the GFT. Specifically, our goal is to express the GFrFT in a hyper-differential form, referred to as HGFrFT

$$\mathcal{F}_\alpha = \exp\left(-i\frac{\pi^2}{2}\alpha(\mathcal{P}^2 + \mathcal{Q}^2)\right), \quad (10)$$

where the hyper-differential operator

$$\mathcal{H}_\alpha = \alpha\pi(\mathcal{P}^2 + \mathcal{Q}^2), \quad (11)$$

depends on the construction of $\mathcal{P}$ and $\mathcal{Q}$.

To preserve the intrinsic connection between vertex and spectral domains, we require that the operators $\mathcal{P}$ and $\mathcal{Q}$ satisfy the graph Fourier duality

$$\mathcal{P} = \mathcal{F}\mathcal{Q}\mathcal{F}^{-1}, \quad \mathcal{Q} = \mathcal{F}\mathcal{P}\mathcal{F}^{-1}. \quad (12)$$

Based on this duality, we propose multiple definitions of $\mathcal{P}$ and $\mathcal{Q}$ using different construction strategies. These methods vary in terms of mathematical rigor and computational efficiency. In the following subsections, we provide both theoretically grounded and practically simplified realizations of the hyper-differential framework.

A. Sylvester Equation Solution

Beyond the duality relationship, we now introduce two additional requirements to ensure that the hyper-differential operator form of HGFrFT aligns with the theoretical structure of graph Fourier analysis.

- *Consistency with GFT:* At the standard Fourier setting $\alpha = 1$, the hyper-differential operator must reduce to the logarithmic form of the GFT

$$\mathcal{H} = \frac{2i}{\pi} \log(\mathcal{F}), \quad (13)$$

where $\mathcal{F}$ is the GFT matrix.

- *Fractional Parameterization:* The operator should scale linearly with the fractional order α , i.e.,

$$\mathcal{H}_\alpha = \alpha\mathcal{H}, \quad (14)$$

which guarantees additivity and homogeneity of the fractional operator.

Combining these with the hyper-differential definition $\mathcal{H}_\alpha = \alpha\pi(\mathcal{P}^2 + \mathcal{Q}^2)$, we obtain the following identity

$$\alpha\pi(\mathcal{P}^2 + \mathcal{Q}^2) = \frac{2i\alpha}{\pi} \log(\mathcal{F}). \quad (15)$$

Substituting the duality relationship $\mathcal{P} = \mathcal{F}\mathcal{Q}\mathcal{F}^{-1}$ into the left-hand side yields

$$\mathcal{F}\mathcal{Q}^2\mathcal{F}^{-1} + \mathcal{Q}^2 = \frac{2i}{\pi^2} \log(\mathcal{F}). \quad (16)$$

To further simplify, we multiply both sides of the equation by $\mathcal{F}$ from the left

$$\mathcal{F}\mathcal{Q}^2 + \mathcal{Q}^2\mathcal{F} = \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}, \quad (17)$$

which is recognized as a standard Sylvester equation in the unknown $\mathcal{Q}^2$. Solving this equation yields a class of operators that not only satisfy graph Fourier duality but also the theoretical conditions of consistency and fractional parameterization, making this method the canonical approach for defining the HGFrFT.

This linear matrix equation admits formal identification with standard Sylvester equation components

$$\mathbf{A} = \mathcal{F}, \quad \mathbf{B} = \mathcal{F}, \quad \mathbf{X} = \mathcal{Q}^2, \quad \mathbf{C} = \frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}. \quad (18)$$

The equation $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$ guarantees a unique solution $\mathbf{X}$ when $\mathbf{A}$ and $-\mathbf{B}$ have disjoint spectra [39], [40], [41]. This spectral non-overlap condition generally holds for connected graphs with non-degenerate Laplacian eigenvalues, ensuring well-posedness of our operator construction. The solution $\mathcal{Q}^2$ enables full recovery of the hyper-differential operator components through the duality relationships. This systematic derivation preserves the theoretical consistency between the hyper-differential formulation and conventional GFrFT definitions while enabling practical computation through linear algebraic solutions. The subsequent sections develop four distinct solution methodologies, each offering unique advantages in computational efficiency and numerical stability.

1) *Vectorized Solution*: An analytical solution to (17) is obtained using the Kronecker product [37]. By vectorizing the equation, we determine $\mathbf{X} = \mathcal{Q}^2$. The Sylvester equation in its vectorized form is expressed as

$$(\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F})\text{vec}(\mathbf{X}) = \text{vec}\left(\frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\right), \quad (19)$$

where $\otimes$ denotes the Kronecker product, and $\text{vec}(\cdot)$ is the vectorization operator. The vectorization follows $\text{vec}(\mathbf{X}) = [\mathbf{X}_{:,0}^\top, \mathbf{X}_{:,1}^\top, \dots, \mathbf{X}_{:,N-1}^\top]^\top$. The Kronecker sum $\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F}$ is assumed to be invertible if $\mathcal{F}$ and $-\mathcal{F}$ have no common eigenvalues. Under this condition, the solution for $\text{vec}(\mathcal{P}^2)$ is given by

$$\text{vec}(\mathcal{Q}^2) = (\mathcal{F} \otimes \mathbf{I} + \mathbf{I} \otimes \mathcal{F})^{-1} \text{vec}\left(\frac{2i}{\pi^2} \log(\mathcal{F})\mathcal{F}\right). \quad (20)$$

Reshaping $\text{vec}(\mathcal{Q}^2)$ into matrix form yields the solution for $\mathcal{Q}^2$, corresponding to the original Sylvester equation. However, directly using (20) to compute the solution leads to very high computational complexity due to the large matrix inversions involved. In contrast, we adopt a more efficient generic solution inspired by [37], which simplifies the computation by using the following expression

$$\mathcal{Q}^2 = \frac{i}{\pi^2} \log(\mathcal{F}). \quad (21)$$

This alternative approach significantly reduces the computational cost while still providing an effective solution.

2) *Schur-Based Solution*: Compute the Schur decomposition of $\mathcal{F} \in \mathbb{C}^{N \times N}$ as

$$\mathcal{F} = \mathbf{E}\mathbf{T}\mathbf{E}^*, \quad (22)$$

where $\mathbf{E}$ is unitary and $\mathbf{T}$ is upper triangular. Define the similarity transforms

$$\mathbf{Y} := \mathbf{E}^* \mathcal{Q}^2 \mathbf{E}, \quad \tilde{\mathbf{C}} := \mathbf{E}^* \frac{2i}{\pi^2} \log(\mathcal{F}) \mathcal{F} \mathbf{E}. \quad (23)$$

Left-multiplying (17) by $\mathbf{E}^*$ and right-multiplying by $\mathbf{E}$ yields the triangular Sylvester equation

$$\mathbf{T}\mathbf{Y} + \mathbf{Y}\mathbf{T} = \tilde{\mathbf{C}}. \quad (24)$$

Since $\mathbf{T}$ is upper triangular, (24) can be solved by backward substitution without forming Kronecker products. The solution is then recovered as [42], [43]

$$\mathcal{Q}^2 = \mathbf{E}\mathbf{Y}\mathbf{E}^*. \quad (25)$$

A sufficient condition for (17) to have a unique solution is $\sigma(\mathcal{F}) \cap (-\sigma(\mathcal{F})) = \emptyset$, which is equivalent to requiring $\lambda_i(\mathcal{F}) + \lambda_j(\mathcal{F}) \neq 0$ for all i and j .

3) *Diagonalized Matrix Representation*: This method assumes that the operator $\mathcal{P}$ shares the eigenvector matrix of $\mathcal{F}$, thereby simplifying the problem by establishing a common spectral basis. The matrix $\mathcal{F}$ is given by $\mathcal{F} = \mathbf{U}^{-1} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^{-1}$, where $\mathbf{\Sigma} = \text{diag}([\sigma_0, \sigma_1, \dots, \sigma_{N-1}])$ is the diagonal matrix of eigenvalues. It is further assumed that the matrix $\mathcal{Q}$ is also diagonalizable using the same eigenvector

TABLE I
RELATIVE RESIDUAL OF DIFFERENT METHODS FOR VARIOUS MATRIX SIZES

Method	16	64	256	512
Vectorized	4.22×10^{-1}	5.73×10^{-1}	/	/
Generic	1.68×10^{-15}	4.06×10^{-15}	1.22×10^{-14}	2.54×10^{-14}
Schur-based	2.71×10^{-15}	4.75×10^{-15}	9.63×10^{-15}	1.55×10^{-14}
Diagonalized	1.02×10^{-15}	2.05×10^{-15}	3.33×10^{-15}	4.83×10^{-15}

matrix $\mathbf{V}$, so that

$$\mathcal{Q} = \mathbf{V}\mathbf{\Theta}\mathbf{V}^{-1}, \quad (26)$$

where $\mathbf{\Theta} = \text{diag}([\theta_0, \theta_1, \dots, \theta_{N-1}])$ is the diagonal matrix of eigenvalues corresponding to $\mathcal{Q}$. Substituting the decompositions of $\mathcal{F}$ and $\mathcal{Q}$ into (16) simplifies the expression by working directly in the eigenvalue space

$$\mathbf{\Theta}^2 \mathbf{\Sigma} + \mathbf{\Sigma} \mathbf{\Theta}^2 = \frac{2i}{\pi^2} \log(\mathbf{\Sigma}) \mathbf{\Sigma}, \quad (27)$$

where $\log(\mathbf{\Sigma}) = \text{diag}([\log \sigma_0, \log \sigma_1, \dots, \log \sigma_{N-1}])$. This reduces the problem to solving the scalar equation for each eigenvalue component

$$\theta_i^2 \sigma_i + \sigma_i \theta_i^2 = \frac{2i}{\pi^2} \log(\sigma_i) \sigma_i, \quad \text{for } i = 0, 1, \dots, N-1. \quad (28)$$

The solution for each eigenvalue θ_i is given by

$$\theta_i = \sqrt{\frac{i}{\pi^2} \log(\sigma_i)}. \quad (29)$$

Finally, the operator $\mathcal{Q}$ is reconstructed by combining the eigenvalue matrix $\mathbf{\Theta}$ with the eigenvector matrix $\mathbf{V}$.

4) *Accuracy and Robustness Analysis*: We evaluate these methods using matrix sizes $N \in \{16, 64, 256, 512\}$. For each N , we generate $\mathcal{F} \in \mathbb{C}^{N \times N}$ by sampling a random complex Gaussian matrix and taking the orthonormal factor from its QR factorization. For the same N , we construct the right-hand side using (18) as $\mathbf{C} = \frac{2i}{\pi^2} \log(\mathcal{F}) \mathcal{F}$. Each method produces an estimate $\mathcal{Q}_{\text{est}}$, and we measure the relative residual using the Frobenius norm as

$$\text{Relative Residual} = \frac{\|\mathcal{F}\mathcal{Q}_{\text{est}} + \mathcal{Q}_{\text{est}}\mathcal{F} - \mathbf{C}\|_F}{\|\mathbf{C}\|_F}. \quad (30)$$

Table I reports the relative residuals for four methods. The Vectorized method shows clear numerical limitations. Its residual stays on the order of 10^{-1} at small sizes, and it does not scale to $N \geq 256$ because the underlying Kronecker system grows rapidly. In contrast, the Generic, Schur-based, and Diagonalized methods achieve residuals near machine precision for all tested sizes. Among these, the Diagonalized method yields the smallest residuals, while the Schur-based solver attains comparable accuracy without requiring explicit diagonalization.

B. Fixed Matrix Simplified Forms

In the previous section, the HGFrFT is constructed through a Sylvester equation, which is derived to ensure consistency with the GFT and adherence to the principle of fractional parameterization. While this method provides strong theoretical guarantees, it also involves matrix logarithms and Sylvester-type solutions, potentially leading to significant computational costs.

In this section, an alternative approach is introduced that is inspired by operator-based fractional transforms in classical signal processing.

Specifically, in traditional discrete FrFT [34], [35], the transformation employs exponential operators based on fixed matrix approximations of differential and coordinate multiplication operators. Motivated by this idea, fixed matrix forms of $\mathcal{P}$ and $\mathcal{Q}$ are defined in the graph setting, and only the graph Fourier duality condition, $\mathcal{P} = \mathcal{F}\mathcal{Q}\mathcal{F}^{-1}$, $\mathcal{Q} = \mathcal{F}\mathcal{P}\mathcal{F}^{-1}$, is enforced. This guarantees that the vertex-frequency domain correspondence is maintained. It is important to note that, unlike the Sylvester-based formulation, these fixed operator definitions do not, in general, satisfy the logarithmic consistency condition or the linear fractional parameterization property. Thus, the resulting transformation cannot be regarded as an exact HGFrFT in the strict sense, but rather as a fast, heuristic approximation with practical utility.

1) *Fixed Form of $\mathcal{Q}$* : We directly construct the coordinate multiplication operator $\mathcal{Q}$ using its definition in traditional signal processing [34], [35]. The operator $\mathcal{Q}$ performs coordinate multiplication and is defined as

$$\mathcal{Q}_{mn} = \begin{cases} \frac{n}{\sqrt{N}}, & \text{for } m = n, \\ 0, & \text{for } m \neq n, \end{cases} \quad (31)$$

where $m, n = 0, 1, \dots, N-1$. Given the definition of $\mathcal{Q}$, the operator $\mathcal{P}$ follows as the graph Fourier dual, as shown in (12), $\mathcal{P} = \mathcal{F}\mathcal{Q}\mathcal{F}^{-1}$, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ represent the GFT matrix and its inverse, respectively.

2) *Fixed Form of $\mathcal{P}$* : Unlike the fixed $\mathcal{Q}$ defined earlier, we directly construct the differential operator $\mathcal{P}$ using its definition in traditional signal processing [34], [35]. The operator $\mathcal{P}^2$ takes the form

$$\mathcal{P}_{mn}^2 = \begin{cases} -\frac{\pi^2}{3h^2} - \frac{1}{6}, & \text{if } m = n \text{ and } N \equiv 0 \pmod{2}, \\ -\frac{\pi^2}{3h^2} - \frac{1}{12}, & \text{if } m = n \text{ and } N \equiv 1 \pmod{2}, \\ -\frac{1}{2}(-1)^{m-n} \csc^2\left(\frac{(m-n)h}{2}\right), & \text{if } m \neq n \text{ and } N \equiv 0 \pmod{2}, \\ -\frac{(-1)^{m-n}}{2} \csc\left(\frac{(m-n)h}{2}\right) \cot\left(\frac{(m-n)h}{2}\right), & \text{if } m \neq n \text{ and } N \equiv 1 \pmod{2}, \end{cases} \quad (32)$$

where $m, n = 0, \dots, N-1$ and $h = 2\pi/N$. Using the definition of $\mathcal{P}$, the operator $\mathcal{Q}$ follows as the graph Fourier dual of $\mathcal{P}$, as given by (12), $\mathcal{Q} = \mathcal{F}\mathcal{P}\mathcal{F}^{-1}$.

IV. NUMERICAL EXPERIMENTS AND ANALYSIS

A. Equivalence Verification

In this section, we first analyze the equivalence between the proposed HGFrFT and the traditional GFrFT. We compare three Sylvester equation-based methods—namely, the Generic, Schur-based, and Diagonalized approaches—with the traditional GFrFT. In theory, these methods should yield identical transformation matrices, that is, $\mathcal{F}_{\alpha}^{\text{Traditional}} = \mathcal{F}_{\alpha}^{\text{HGFrFT}}$. Note that

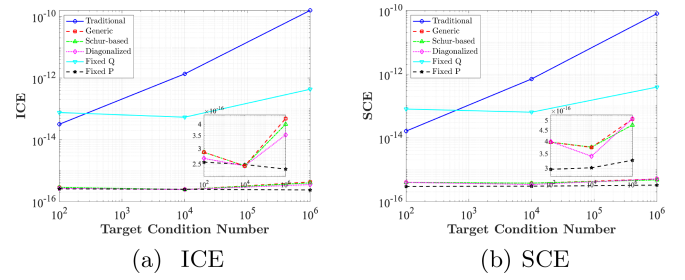


Fig. 1. Stability analysis under an ill-conditioned eigen-basis with $N = 64$.

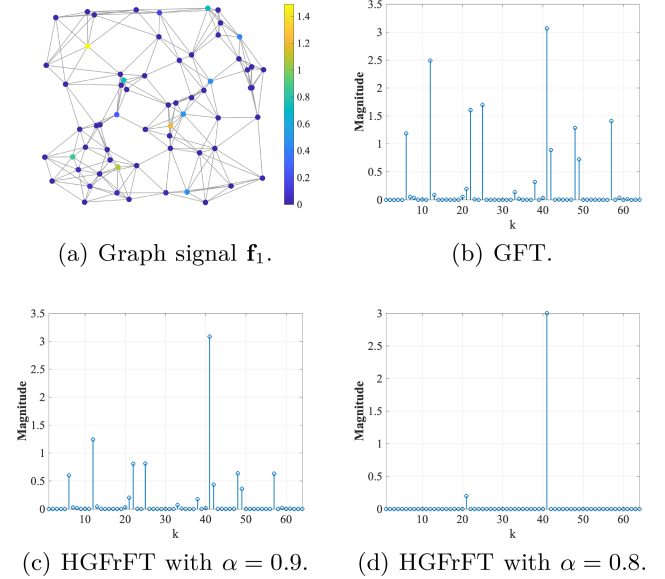


Fig. 2. The GFT and HGFrFT of the graph signal $\mathbf{f}_1$.

the Vectorized method is theoretically equivalent to the Generic method, but it is excluded from this comparison due to its high computational complexity. The Fixed $\mathcal{Q}$ and Fixed $\mathcal{P}$ methods are omitted because they do not guarantee full theoretical equivalence.

To quantitatively assess the equivalence, the normalized mean squared error (NMSE) between the transformation matrices is used

$$\text{NMSE} = \frac{\|\mathcal{F}_{\alpha}^{\text{Traditional}} - \mathcal{F}_{\alpha}^{\text{HGFrFT}}\|_F^2}{\|\mathcal{F}_{\alpha}^{\text{Traditional}}\|_F^2}, \quad (33)$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

The evaluation is conducted using a sensor network graph with 64 vertices, whose structure is shown in Fig. 2(a). This evaluation is conducted using several specific graph shift operators, including the adjacency matrix, Laplacian matrix, row-normalized adjacency matrix, symmetric normalized adjacency matrix, and normalized Laplacian matrix. For each operator, we test across three representative fractional parameters: $\alpha = 0.6$, $\alpha = 0.9$, and $\alpha = 1.2$, to ensure comprehensive coverage. As shown in Table II, all three methods produce the identical NMSE values, which remain close to machine precision for all test cases. These results indicate that the implementations are numerically consistent and support the theoretical equivalence between the proposed HGFrFT formulations and the traditional GFrFT.

TABLE II
NMSE BETWEEN HGFrFT METHODS AND TRADITIONAL GFrFT

Shift Operator	Generic			Schur-based			Diagonalized		
	$\alpha = 0.6$	$\alpha = 0.9$	$\alpha = 1.2$	$\alpha = 0.6$	$\alpha = 0.9$	$\alpha = 1.2$	$\alpha = 0.6$	$\alpha = 0.9$	$\alpha = 1.2$
Adjacency	2.16×10^{-17}	1.47×10^{-17}	1.29×10^{-17}	2.16×10^{-17}	1.47×10^{-17}	1.29×10^{-17}	2.16×10^{-17}	1.47×10^{-17}	1.29×10^{-17}
Laplacian	3.59×10^{-13}	3.02×10^{-18}	2.11×10^{-18}	3.59×10^{-13}	3.02×10^{-18}	2.11×10^{-18}	3.59×10^{-13}	3.02×10^{-18}	2.11×10^{-18}
Normalized Adjacency	1.12×10^{-15}	7.56×10^{-16}	6.77×10^{-16}	1.12×10^{-15}	7.56×10^{-16}	6.77×10^{-16}	1.12×10^{-15}	7.56×10^{-16}	6.77×10^{-16}
Normalized Laplacian	3.84×10^{-12}	1.32×10^{-16}	1.14×10^{-16}	3.84×10^{-12}	1.32×10^{-16}	1.14×10^{-16}	3.84×10^{-12}	1.32×10^{-16}	1.14×10^{-16}

B. Computational Efficiency

In this section, we analyze the computational efficiency of the proposed solutions for implementing the HGFrFT.

- Vectorized Solution:** The computational complexity mainly arises from solving the Kronecker product system, which has a complexity of $O(N^6)$, and computing the matrix exponential, which contributes $O(N^3)$. Therefore, the overall complexity is dominated by $O(N^6)$.
- Generic Solution:** In contrast, the generic solution simplifies the computation by avoiding the expensive Kronecker product. It has a complexity of $O(N^3)$, as it only involves matrix logarithm and multiplication, making it more efficient for large matrices.
- Schur-based Solution:** The Schur decomposition and solving the triangular Sylvester equation via back substitution each require $O(N^3)$. Matrix multiplications for similarity transforms and the final matrix exponential also contribute $O(N^3)$, yielding overall $O(N^3)$ complexity.
- Diagonalized Representation:** By leveraging matrix diagonalization, this method significantly reduces computational overhead. The diagonalization step requires $O(N^3)$, and subsequent computations in the eigenvalue space, including the matrix exponential, remain at $O(N^3)$. As a result, the total complexity is $O(N^3)$, making it one of the most efficient solutions.
- Fixed Form of $\mathcal{Q}$:** Constructing $\mathcal{Q}$ in a fixed diagonal form incurs only $O(N)$ cost, while the dominant complexity arises from computing $\mathcal{F}_\alpha$, which requires $O(N^3)$. This method retains a total complexity of $O(N^3)$ but benefits from a simplified matrix structure.
- Fixed Form of $\mathcal{P}$:** The construction of $\mathcal{P}$ involves $O(N^2)$ operations, slightly higher than that of $\mathcal{Q}$. However, the final matrix exponential still dominates with $O(N^3)$ complexity, leading to an overall computational cost of $O(N^3)$. This method balances efficiency and flexibility.

To evaluate the computational efficiency of the proposed methods, we performed simulations on matrices of various sizes and compared the running times. The test matrices are generated as random complex matrices with entries following a standard complex normal distribution. The results are summarized in Table III. The Generic Solution is efficient across all tested sizes, significantly reducing computation time, especially for larger matrices like $N = 1024$, by avoiding the Kronecker product and relying on matrix logarithm and multiplication. The Schur-based Solution achieves comparable performance to Generic, while the Diagonalized Representation and both Fixed $\mathcal{Q}$ and Fixed $\mathcal{P}$ methods stand out for their low computational cost. The

TABLE III
RUNNING TIME OF DIFFERENT METHODS FOR VARIOUS MATRIX SIZES (IN SECONDS)

Method	16	64	128	256	512	1024
Generic	4.7735	0.2430	0.1882	0.7296	3.4543	15.0319
Schur-based	0.0222	0.0710	0.1485	0.6296	3.4734	14.9765
Diagonalized	0.0095	0.0590	0.1505	0.5989	3.1161	13.3123
Fixed $\mathcal{Q}$	0.0040	0.0205	0.0293	0.1116	0.8225	4.6394
Fixed $\mathcal{P}$	0.0224	0.0354	0.0286	0.1012	0.5145	4.1165
Traditional	0.0022	0.0150	0.0427	0.1898	1.1014	5.0289

TABLE IV
MULTI-QUERY AMORTIZATION RESULTS ON THE RING GRAPH WITH $N = 64$

Method	Direct total (s)	Reuse-enabled total (s)	Speedup factor
Traditional	0.4625	0.3889	1.19×
Generic	0.9953	0.3620	2.75×
Schur-based	0.7362	0.2546	2.89×
Diagonalized	0.8845	0.2665	3.34×
Fixed $\mathcal{Q}$	0.4260	0.2641	1.61×
Fixed $\mathcal{P}$	0.3506	0.2293	1.53×

proposed HGFrFT implementations provide computationally efficient alternatives to the traditional GFrFT.

In addition to the single-run timings in Table III, we evaluate efficiency when multiple fractional orders are queried on the same graph. In many applications, one sweeps a range of fractional orders α and applies the corresponding transforms to a batch of signals. The key question is whether a method can reuse graph-specific quantities computed once, so that forming $\mathcal{F}_\alpha$ for different values of α becomes less expensive. We perform a multi-query amortization test on the RING graph with $N = 64$. We generate 50 random input signals and evaluate 15 fractional orders, for a total of 750 transform applications. For each method, we consider two implementations. The direct implementation constructs the transform matrix independently for each α and then applies it to all signals. The reuse-enabled implementation precomputes graph-specific factors once, such as decompositions or operator factors, and reuses them when forming the transform matrices for different values of α . We report the total runtime over all 750 applications and compute the speedup as the ratio between the direct and reuse-enabled runtimes.

The results in Table IV show that reuse consistently improves the efficiency of the proposed HGFrFT variants. The Diagonalized method delivers the largest reduction in total runtime with a 3.34× speedup, followed by the Schur-based method with 2.89× and the Generic method with 2.75×. Fixed $\mathcal{Q}$ and Fixed $\mathcal{P}$ also improve with speedups of 1.61× and 1.53×, respectively, which is expected because their per query construction is already inexpensive. By comparison, the Traditional baseline

gains only $1.19\times$, suggesting limited reuse beyond caching an eigendecomposition.

Although the Traditional method is competitive when constructing a single transform matrix, the HGFrFT implementations are more suitable for repeated fractional queries because they support a clear precompute once and reuse many times workflow. This makes the proposed operator-based formulations more efficient when multiple values of α are needed on the same graph.

C. Stability Analysis

We assess numerical robustness when the eigenvector basis is poorly conditioned. We start from the representation $\mathcal{F} = \mathbf{V}\Sigma\mathbf{V}^{-1}$ and generate a set of matrices $\mathcal{F}$ by controlling the condition number of $\mathbf{V}$ while keeping the spectrum Σ bounded. We fix $N = 64$ and test condition numbers of 10^2 , 10^4 , and 10^6 . For each case, we compute the HGFrFT operator using several implementations, including Traditional, Generic, Schur-based, Diagonalized, Fixed $\mathcal{Q}$, and Fixed $\mathcal{P}$. We then check whether the computed operators satisfy key algebraic consistency conditions.

We report two consistency errors measured with the Frobenius norm. The inverse-consistency error (ICE) evaluates how close $\mathcal{F}_\alpha\mathcal{F}_{-\alpha}$ is to the identity matrix:

$$\text{ICE}(\alpha) := \frac{\|\mathcal{F}_\alpha\mathcal{F}_{-\alpha} - \mathbf{I}\|_F}{\|\mathbf{I}\|_F}. \quad (34)$$

The semigroup-consistency error (SCE) evaluates how well the composition rule holds by comparing $\mathcal{F}_\alpha\mathcal{F}_\beta$ with $\mathcal{F}_{\alpha+\beta}$:

$$\text{SCE}(\alpha, \beta) := \frac{\|\mathcal{F}_\alpha\mathcal{F}_\beta - \mathcal{F}_{\alpha+\beta}\|_F}{\|\mathcal{F}_{\alpha+\beta}\|_F}. \quad (35)$$

As shown in Fig. 1, the Traditional implementation becomes increasingly sensitive as the condition number of $\mathbf{V}$ grows, and both ICE and SCE increase noticeably. This trend is consistent with the amplification of rounding errors induced by repeated similarity transforms involving $\mathbf{V}^{-1}$ when the eigen-basis is poorly conditioned. Fixed $\mathcal{Q}$ exhibits a comparable sensitivity pattern. In contrast, Generic, Schur-based, and Diagonalized remain close to machine precision across the entire range of eigen-basis conditioning, indicating improved robustness in maintaining the operator consistency constraints.

D. Graph Signal Examples

To demonstrate the application of the HGFrFT in graph signal transformation, we analyze two distinct graph scenarios. First, we examine a sensor network graph with 64 vertices. The graph signal is given by

$$\mathbf{f}_1 = 0.2\Re(\varsigma_{20}(n)) + 3\Re(\varsigma_{40}(n)), \quad (36)$$

where $\Re(\cdot)$ denotes the real part, and $\varsigma_k(n)$ for $k = 20$ and $k = 40$ is derived from the HGFrFT of the sensor network graph. The HGFrFT of this graph, along with its corresponding transformation outputs, is presented in Fig. 2(a). Fig. 2(b)–(c) collectively provide a comprehensive visualization of the transformation outcomes across various parameter settings. Next, we

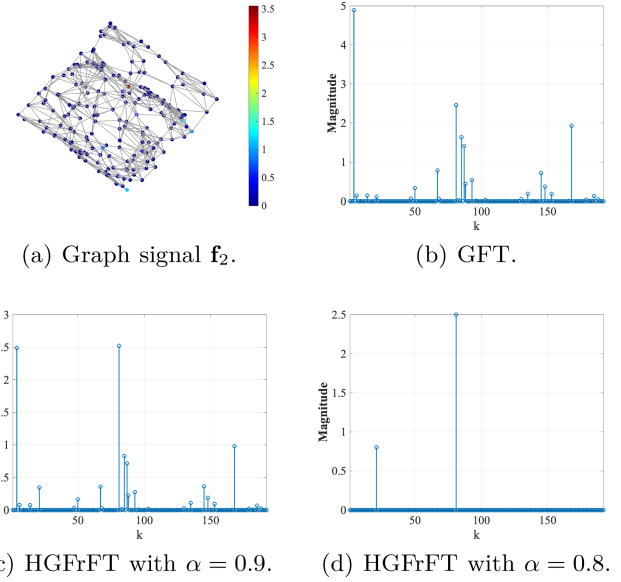


Fig. 3. The GFT and HGFrFT of the graph signal $\mathbf{f}_2$.

analyze a Swiss roll graph with 192 vertices. The graph signal is given by

$$\mathbf{f}_2 = 0.8\Re(\varsigma_{20}(n)) + 2.5\Re(\varsigma_{80}(n)), \quad (37)$$

where $\varsigma_k(n)$ for $k = 20$ and $k = 80$ is derived in the same manner from the HGFrFT of the Swiss roll graph. Fig. 3(a) presents the HGFrFT of the Swiss roll graph, while Fig. 3(b)–(c) provide a comprehensive visualization of the transformation results across various parameter configurations.

In both cases, the signals are intentionally constructed as linear combinations of specific HGFrFT basis vectors. This design choice allows for a clear and intuitive demonstration of how the fractional parameter α influences the sparsity and alignment of signal components in the transformed domain. Since real-world graph signals often lack analytical forms and exhibit highly irregular structures, it is challenging to isolate the effect of the fractional parameter through natural signals alone. Therefore, we adopt these illustrative examples to better visualize the mechanism and flexibility of the HGFrFT in adapting to signal structures. By setting $\alpha = 0.8$, we observe that the HGFrFT achieves compact and informative spectral representations, allowing for precise identification of the frequency components in the graph signals. This demonstrates its potential in capturing relevant features through fractional tuning. The flexibility of the HGFrFT parameter significantly enhances spectral analysis, yielding a more accurate and insightful representation of graph signals. This constructed setup provides a focused view of the transform's behavior, while the broader applicability and performance of HGFrFT on real-world signals are further validated in Section VII through ECG classification tasks.

V. PARAMETER OPTIMIZATION

The fractional parameter α controls the order of the transform and thus affects the representation of graph signals. In this section, we analyze the problem of parameter optimization.

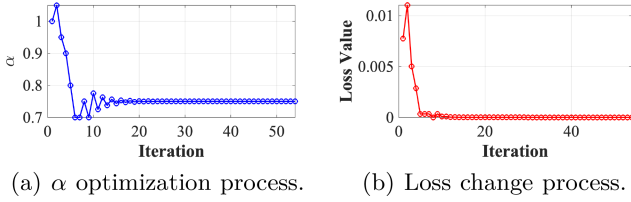


Fig. 4. Representative convergence behavior showing α optimization and loss function reduction using the Diagonalized method.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS IN PARAMETER OPTIMIZATION

Method	Iterations	Time (s)	Final Error	α_{est}
Vectorized	54	0.6277	3.08×10^{-33}	0.75
Generic	54	0.1670	7.70×10^{-34}	0.75
Schur-based	54	0.1085	3.08×10^{-33}	0.75
Diagonalized	54	0.1968	3.08×10^{-33}	0.75
Fixed $\mathcal{Q}$	54	0.1417	3.20×10^{-30}	0.75
Fixed $\mathcal{P}$	54	0.1948	2.74×10^{-48}	0.75

Given a target transformed signal $\hat{\mathbf{f}}_{\alpha_{\text{orig}}}$ and an initial signal $\mathbf{f}$, the objective is to determine the optimal parameter α_{est} that minimizes the loss function

$$\mathcal{R}(\alpha_{\text{est}}) = \|\mathbf{f}_{\alpha_{\text{est}}} - \mathbf{f}_{\alpha_{\text{orig}}}\|^2, \quad (38)$$

where $\mathbf{f}_{\alpha} = \mathcal{F}_{\alpha}\mathbf{f}$. The optimization is performed using the Nelder-Mead simplex method [44], a gradient-free algorithm that iteratively refines α_{est} through reflection, expansion, and contraction operations. This approach is robust for low-dimensional parameter spaces and eliminates the need for explicit gradient computations, making it suitable for optimizing the HGFrFT parameter α .

We evaluate the optimization performance using a sensor network graph with 24 vertices generated via the GSP Toolbox. The test signal $\mathbf{f} \in \mathbb{R}^{24}$ follows a uniform random distribution $\mathcal{U}(-1, 1)$. Starting from $\alpha_0 = 1.0$, the algorithm aims to recover the target parameter $\alpha_{\text{orig}} = 0.75$ with a convergence tolerance of $\epsilon = 10^{-8}$. The optimization trajectories demonstrate consistent behavior across all methods due to identical algorithmic configurations. Fig. 4 illustrates the convergence process from the initial state to the target value using the Diagonalized method.

As shown in Table V, all methods converge to the target parameter within 54 iterations and achieve high numerical precision. Among them, the Schur-based method is the fastest, followed by Fixed $\mathcal{Q}$ and Generic. The methods demonstrate exceptional numerical precision with final error values below 10^{-30} . In practice, α can be selected by sweeping a small candidate set and choosing the value that yields the best task-specific performance.

VI. VERTEX-FREQUENCY ANALYSIS AND OPTIMAL WINDOW DESIGN

A. Vertex-Frequency Analysis

To further enhance its applicability, we examine methods related to vertex-frequency analysis [45], [46]. For HGFrFT in

(10), we express it as

$$\mathcal{F}_{\alpha} = \exp\left(-i\frac{\pi^2}{2}\alpha(\mathcal{P}^2 + \mathcal{Q}^2)\right) = [\varsigma_0, \varsigma_1, \dots, \varsigma_{N-1}]. \quad (39)$$

Then, the HGFrFT is given by

$$\hat{f}_{\alpha}(k) = \sum_{n=0}^{N-1} f(n)\varsigma_k(n). \quad (40)$$

The primary objective of vertex-frequency analysis is to characterize the distribution of frequencies across individual vertices in a graph. This is accomplished by applying a window function, enabling the derivation of the vertex-frequency distribution $\mathcal{V}_{\alpha}(n, k)$. In practice, window functions are generally categorized into two types: vertex-domain functions and frequency-domain functions.

First, we analyze the construction of the window function in the vertex domain. The shortest path length between two vertices, denoted as $\tau_{n,m}$, is computed using the graph's adjacency matrix $\mathbf{A}$. Using the computed shortest path length $\tau_{n,m}$, we define a localized window function $g(\tau_{n,m})$, where $g(\cdot)$ represents a standard window function commonly used in classical signal processing. This localized window function serves as a foundation for analyzing signals on graphs by emphasizing local structures defined through vertex neighborhoods. This relationship is characterized using the path adjacency matrix, denoted as $\mathbf{A}_{\tau_{n,m}}$, which is defined as

$$\mathbf{A}_{\tau_{n,m}} = \begin{cases} \mathbf{A}, & \tau_{n,m} = 1, \\ (\mathbf{A} \odot \mathbf{A}_{\tau_{n,m}-1}) \circ (\mathbf{1} - \mathbf{A}_{\tau_{n,m}-1}) \circ (\mathbf{1} - \mathbf{I}), & \tau_{n,m} \geq 2. \end{cases}$$

Here, $\mathbf{A}_{\tau_{n,m}}$ encodes the connectivity information for paths of length $\tau_{n,m}$. The Boolean matrix product $\odot$ identifies connectivity for paths of a given length, while the Hadamard product $\circ$ isolates vertices connected by paths of exactly $\tau_{n,m}$. Specifically, the m -th row of $(\mathbf{A} \odot \mathbf{A}_{\tau_{n,m}-1})$ represents all vertices that are reachable from vertex m through paths of length $\tau_{n,m}$ or shorter. The term $(\mathbf{1} - \mathbf{A}_{\tau_{n,m}-1})$ eliminates connections established by shorter paths, while $(\mathbf{1} - \mathbf{I})$ removes diagonal elements, thereby preventing self-loops. This formulation offers a precise framework for computing graph connectivity based on specific path lengths, facilitating detailed vertex-frequency analysis [47].

For a given window function $\mathbf{g} = [g(0), g(1), \dots, g(N-1)]^T \in \mathbb{R}^N$, the localized window function, denoted by $\mathbf{g}_{\text{loc}}$, is defined with a graph window width of η and expressed as

$$\mathbf{g}_{\text{loc}} = g(0)\mathbf{I} + g(1)\mathbf{A}_1 + \dots + g(\eta-1)\mathbf{A}_{\eta-1}. \quad (41)$$

The vertex-frequency distribution using the HGFrFT for a graph signal $f(n)$ is computed as

$$\mathcal{V}_{\alpha}f(m, k) = \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} f(n)g_{\text{loc}}(n, m)\varsigma_k(n). \quad (42)$$

Consider a path graph with uniform edge weights and the following two signals

$$f_3(n) = \sin(90\pi n/N), \quad 0 \leq n \leq 255, \quad (43)$$

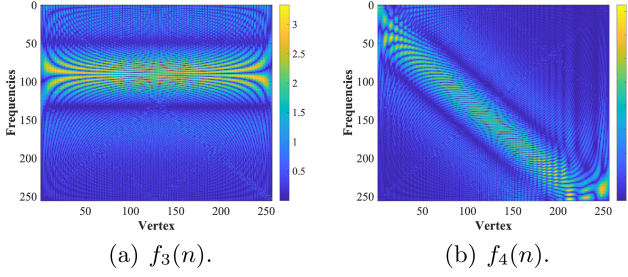


Fig. 5. The vertex-frequency distribution of $f_3(n)$ and $f_4(n)$ using the HGFrFT with a parameter $\alpha = 0.95$.

$$f_4(n) = \sin((25n + 0.5n^2)\pi/N), \quad 0 \leq n \leq 255, \quad (44)$$

representing a sampled sinusoidal waveform and a chirp signal, respectively. Fig. 5 illustrates the vertex-frequency distribution of $f_3(n)$ and $f_4(n)$ using the HGFrFT with a parameter $\alpha = 0.95$. This example uses the path graph as it offers a simple and interpretable topology where the variation of local frequency content along vertices can be clearly observed. Such structure facilitates intuitive visualization of vertex-frequency distributions, especially for non-stationary signals. In this section, we employ the Diagonalized method, which remains the default approach unless stated otherwise.

Additionally, we analyze the structure of the window function in the spectral domain. Let $f(n)$ and $h(n)$ be two signals on a graph, with their respective HGFrFTs given by $\hat{f}_\alpha$ and $\hat{h}_\alpha$. The HGFrFT convolution, denoted by $*_\alpha$, is formulated as

$$(f *_\alpha h)(n) = \mathcal{F}_\alpha^{-1} \{ \hat{f}_\alpha(k) \hat{h}_\alpha(k) \} = \sum_{k=0}^{N-1} \hat{f}_\alpha(k) \hat{h}_\alpha(k) \varsigma_k^*(n). \quad (45)$$

For any signal $g(n)$ on a graph, let $\delta_m(n)$ denote a delta function centered at vertex $m \in \{0, 1, \dots, N-1\}$, defined as $\delta_m(n) = \delta(n - m)$. The vertex-frequency representation of the delta function using HGFrFT is given by

$$\Delta_m(k) = \mathcal{F}_\alpha \{ \delta_m(n) \} = \sum_{n=0}^{N-1} \delta_m(n) \varsigma_k(n) = \varsigma_k(m). \quad (46)$$

The HGFrFT-based translation operator T_m^α is defined as

$$T_m^\alpha g(n) = (g *_\alpha \delta_m)(n) = \sum_{k=0}^{N-1} \hat{g}_\alpha(k) \varsigma_k(m) \varsigma_k^*(n). \quad (47)$$

Consequently, the vertex-frequency representation of the graph signal $f(n)$ using HGFrFT is formulated as

$$\mathcal{V}_\alpha f(m, k) = \sum_{n=0}^{N-1} f(n) T_m^\alpha g^*(n) \varsigma_k(n), \quad (48)$$

where $T_m^\alpha g(n)$ follows the definition in (47). Applying the graph window function $T_m^\alpha g(n)$ to $f(n)$ at each vertex and summing over all vertices n preserves the overlap-add property. The inverse vertex-frequency representation using HGFrFT is expressed as $f(n) = \frac{1}{\sum_{i=0}^{N-1} T_m^\alpha g(n)} \sum_{k=0}^{N-1} \mathcal{V}_\alpha f(m, k) \varsigma_k^*(n)$.

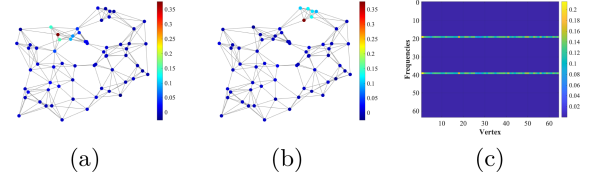


Fig. 6. Vertex-frequency analysis based on HGFrFT (Diagonalized method) of the sensor network graph with 64 vertices. (a) Vertex 20. (b) Vertex 40. (c) Vertex-frequency distribution of the graph signal $f_5(n)$.

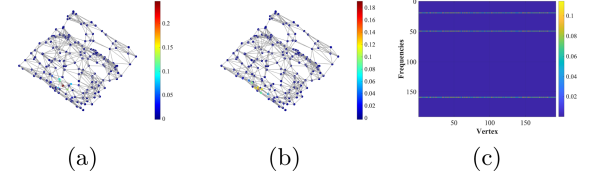


Fig. 7. Vertex-frequency analysis based on HGFrFT (Diagonalized method) of the Swiss roll graph with 192 vertices. (a) Vertex 90. (b) Vertex 150. (c) Vertex-frequency distribution of the graph signal $f_6(n)$.

We analyze a sensor network graph with 64 vertices. The spectral domain window is defined as $G(k) = e^{-400\lambda_k}$ with $\alpha = 0.9$. Local vertex-frequency analysis is performed at vertices 20 (Fig. 6(a)) and 40 (Fig. 6(b)). The graph signal $f_5(n)$ on the sensor network graph is given by

$$f_5(n) = \Re(\varsigma_{20}(n)) + \Re(\varsigma_{40}(n)), \quad (49)$$

where $\varsigma_k(n)$ represents components derived from the graph fractional Fourier matrix. Fig. 6(c) presents the vertex-frequency distribution of $f_5(n)$ using HGFrFT. For the Swiss roll graph with 192 vertices, vertex-frequency analysis is conducted at vertices 90 (Fig. 7(a)) and 150 (Fig. 7(b)). The graph signal $f_6(n)$ on the Swiss roll graph is defined as

$$f_6(n) = \Re(\varsigma_{20}(n)) + \Re(\varsigma_{50}(n)) + \Re(\varsigma_{160}(n)), \quad (50)$$

where $\varsigma_k(n)$ is derived from the HGFrFT of the graph. With the spectral window set to $G(k) = e^{-3500\lambda_k}$ and $\alpha = 0.9$, Fig. 7(c) illustrates the vertex-frequency distribution of $f_6(n)$ using HGFrFT. These two signals are constructed using selected basis components of the HGFrFT in order to demonstrate how the vertex-frequency representation behaves on complex and irregular graph structures.

B. Optimal Window Design

In this section, we analyze the optimal window function. To construct a convolution operator using the HGFrFT, we first recall that the operator defined in (45) is formulated as

$$\mathbf{f} *_\alpha \mathbf{g} = \mathcal{F}_\alpha^{-1} (\mathcal{F}_\alpha \mathbf{f} \odot \mathcal{F}_\alpha \mathbf{g}), \quad (51)$$

where $\odot$ denotes element-wise multiplication. In GSP, this operation is represented as

$$\mathbf{f} *_\alpha \mathbf{g} = \mathcal{F}_\alpha^{-1} \mathbf{G}(\Lambda_\alpha) \mathcal{F}_\alpha \mathbf{f} = \mathbf{G}(\mathcal{L}_\alpha) \mathbf{f}, \quad (52)$$

where the diagonal matrix $\mathbf{G}(\Lambda_\alpha) = \text{diag}(\hat{\mathbf{g}}_\alpha)$ represents the spectral characteristics of the chosen window function.

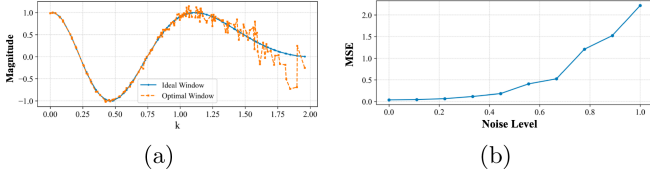


Fig. 8. (a) Comparison of the ideal window (blue), the proposed optimal window (orange). (b) The relationship between noise level ϵ and the MSE.

Consider a random input signal $\mathbf{f} \in \mathbb{R}^N$, and the goal is to estimate the signal $\mathbf{h} = \mathbf{H}\mathbf{f} + \mathbf{n}$, where $\mathbf{H}$ is a known linear operator and $\mathbf{n}$ represents additive noise. The objective is to design a window $\mathbf{G}(\mathcal{L}_\alpha)$ such that the estimated output

$$\tilde{\mathbf{h}} = \mathbf{G}(\mathcal{L}_\alpha)\mathbf{f}. \quad (53)$$

minimizes the mean-squared error with respect to $\mathbf{h}$. The loss function is defined as

$$\mathcal{R}(\mathbf{G}) = \frac{1}{N} \|\tilde{\mathbf{h}} - \mathbf{h}\|_F^2 = \frac{1}{N} \|\mathbf{G}(\mathcal{L}_\alpha)\mathbf{f} - \mathbf{h}\|_F^2, \quad (54)$$

which leads to the following optimization problem

$$\min_{\mathbf{G}} \frac{1}{N} \|\mathbf{G}(\mathcal{L}_\alpha)\mathbf{f} - \mathbf{h}\|_F^2. \quad (55)$$

For a given α , the optimal window in the spectral domain is obtained by transforming $\mathbf{f}$ and $\mathbf{h}$ using $\mathcal{F}_\alpha$. The squared loss $\hat{\mathcal{R}}$ in the spectral domain is expressed as

$$\hat{\mathcal{R}}(\mathbf{G}) = \frac{1}{N} \|\mathbf{G}(\Lambda)\mathcal{F}_\alpha\mathbf{f} - \mathcal{F}_\alpha\mathbf{h}\|_F^2. \quad (56)$$

Differentiating $\hat{\mathcal{R}}$ with respect to $\mathbf{G}$, we have

$$(\mathcal{F}_\alpha\mathbf{f} \odot \mathcal{F}_\alpha\mathbf{f})\mathbf{1}_N \odot \mathbf{G}^{\text{opt}} = (\mathcal{F}_\alpha\mathbf{f} \odot \mathcal{F}_\alpha\mathbf{h})\mathbf{1}_N. \quad (57)$$

Therefore, the optimal window in the fractional spectral domain is given by the ratio of the transformed output to the transformed input.

We analyze a K-NN graph with $N = 144$ vertices, where each vertex is connected to 5 neighbors. The graph-frequency content of the signal follows a normal distribution

$$\hat{f}_7(k) \sim \mathcal{N}(e^{-1.5\lambda_k}, (e^{-1.5\lambda_k}/2)^2). \quad (58)$$

The matrix-valued graph signal is constructed as

$$\mathbf{f}_7 = \mathcal{F}_\alpha^{-1}\hat{\mathbf{f}}_7 \in \mathbb{R}^{144 \times 200}. \quad (59)$$

For the HGFrFT, we set $\alpha = 0.9$. The ideal window function in the spectral domain is given by

$$H(k) = \sin(2(\lambda_k - 2)^2), \quad k = 0, \dots, N-1. \quad (60)$$

The noisy measurement follows the model

$$\mathbf{h} = \mathcal{F}_\alpha^{-1}\mathbf{G}(\Lambda)\mathcal{F}_\alpha\mathbf{f}_7 + \mathbf{n}, \quad (61)$$

where the noise term follows $\mathbf{n}(n) \sim \mathcal{N}(0, \epsilon^2)$, $n \in \{1, \dots, N\}$. Fig. 8(a) compares the optimal window with the ideal one.

To assess the robustness of the proposed filtering approach, we introduce varying noise levels $\epsilon \in [0, 1]$ into the measurements

and reconstruct the original signal. Fig. 8(b) illustrates the relationship between the noise level ϵ and the mean squared error (MSE). As expected, the MSE increases as the noise level rises. At low noise levels ($\epsilon < 0.2$), the MSE remains close to zero, indicating effective signal reconstruction. For moderate noise levels ($\epsilon \in [0.2, 0.6]$), the MSE increases gradually, reflecting a decrease in noise suppression. At high noise levels ($\epsilon > 0.6$), the MSE increases sharply, indicating that noise severely affects the reconstruction process.

VII. APPLICATION OF HGFrFT IN ECG CLASSIFICATION

In this section, we explore the application of HGFrFT to ECG signals. Since ECG signals are naturally represented as time series, modeling them as graph-structured data enables the capture of intrinsic geometric relationships among multiple heartbeat samples [48], [49]. We use the MIT-BIH Arrhythmia Database, available at <https://archive.physionet.org/cgi-bin/atm/ATM>, which offers comprehensive data spanning a wide range of categories. This database is annotated by authoritative experts and is widely used in arrhythmia research [50], [51].

A. Data Preparation and Graph Construction

For the arrhythmia detection task, we select the following four heartbeat types: normal beat (N), left bundle branch block beat (LBBB), right bundle branch block beat (RBBB), and premature ventricular beat (PVC). We select 1000 samples for each type of heartbeat, with each heartbeat having a length of 301 samples (150 samples before and after the R-wave), denoted as $\mathbf{X}_N, \mathbf{X}_{LBBB}, \mathbf{X}_{RBBB}, \mathbf{X}_{PVC} \in \mathbb{R}^{301 \times 1000}$.

To convert these time-series signals into a graph structure suitable for processing with HGFrFT, we construct an adjacency matrix $\mathbf{A} \in \mathbb{R}^{301 \times 301}$ that encodes the similarity relationships among the ECG samples. We learn $\mathbf{A}$ by minimizing the reconstruction loss

$$\mathcal{R} = \|\mathbf{X} - \mathbf{X}\mathbf{A}\|_F^2, \quad (62)$$

where $\mathbf{X} = [\mathbf{X}_N, \mathbf{X}_{LBBB}, \mathbf{X}_{RBBB}, \mathbf{X}_{PVC}]^\top \in \mathbb{R}^{4000 \times 301}$ is the combined data matrix. This formulation encourages each sample to be approximated by a linear combination of other samples, revealing underlying signal relationships. The optimization uses gradient descent with

$$\frac{\partial \mathcal{R}}{\partial \mathbf{A}} = -2(\mathbf{X}^\top \mathbf{X} - \mathbf{A}\mathbf{X}^\top \mathbf{X}), \quad (63)$$

and update rule $\mathbf{A} \leftarrow \mathbf{A} - \eta \frac{\partial \mathcal{R}}{\partial \mathbf{A}}$. Key constraints enforced are: (1) zero diagonal elements to eliminate self-loops; (2) normalized row and column sums: $\mathbf{A}\mathbf{1} = \mathbf{1}$ and $\mathbf{A}^\top \mathbf{1} = \mathbf{1}$. Fig. 9(a) shows the resulting adjacency matrix.

B. Feature Analysis

Building upon the established equivalence of the proposed methods, we next conduct ECG feature analysis utilizing the Diagonalized approach with $\alpha = 0.95$. Fig. 9(b) presents the transformed result for a normal heartbeat as an example. The four heartbeat categories are then transformed separately, with the results presented in Fig. 10. The boxplot in Fig. 10(a)

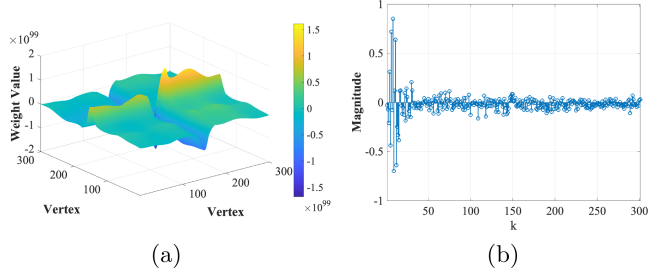


Fig. 9. (a) The obtained adjacency matrix. (b) The HGFrFT of a normal heartbeat.

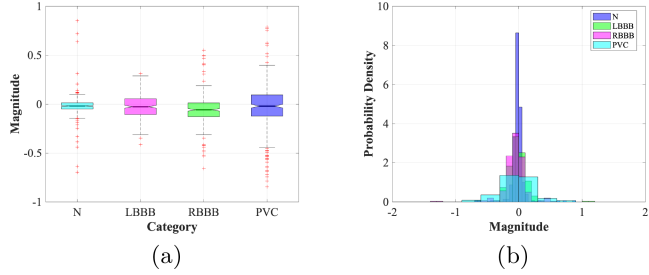


Fig. 10. Feature distribution of ECG signals after transformation using $\mathcal{F}_\alpha$ (Diagonalized). (a) Boxplot of transformed heartbeat data (b) Histogram of transformed heartbeat data distribution.

illustrates that N and RBBB exhibit compact interquartile ranges (IQRs), indicating feature concentration, whereas PVC and LBBB display wider IQRs with more outliers, suggesting greater variability. The histogram in Fig. 10(b) further highlights feature separability, with N displaying a sharply peaked distribution, whereas PVC and LBBB exhibit broader spreads, indicating dispersed feature distributions. These distinct statistical patterns validate the effectiveness of HGFrFT in enhancing feature discrimination, reinforcing its potential for automated heartbeat classification and anomaly detection.

C. Classification Performance Evaluation

In this section, we apply the extracted features to practical classification tasks based on the preceding feature analysis. The dataset is divided into training (70%) and test (30%) sets.

We first evaluate classification performance using features extracted by the standard GFT and by the proposed HGFrFT framework. Earlier results show that HGFrFT constructions based on the Sylvester equation produce the same fractional operator $\mathcal{F}_\alpha$ as the Traditional GFrFT, up to numerical precision, when they start from the same matrix $\mathcal{F}$. For this downstream task, we therefore use the Diagonalized realization as a representative solver-based HGFrFT implementation. We also evaluate two fixed-operator variants, Fixed $\mathcal{Q}$ and Fixed $\mathcal{P}$. All fractional methods use $\alpha = 0.9$. Classification is performed using a neural network with 4 hidden neurons, and each method and shift-operator setting is evaluated over 1000 independent trials.

As shown in Table VI, using a fractional order improves classification accuracy over the standard GFT for several shift operators. The Diagonalized HGFrFT consistently attains strong performance. Both Fixed $\mathcal{Q}$ and Fixed $\mathcal{P}$ also achieve high

TABLE VI
CLASSIFICATION ACCURACY (%) ACROSS DIFFERENT METHODS AND GRAPH SHIFT OPERATORS

Shift Operator	GFT	Diagonalized	Fixed $\mathcal{Q}$	Fixed $\mathcal{P}$
Adjacency	91.96	97.32	97.25	97.32
Laplacian	96.21	98.99	96.68	91.92
Normalized Adjacency	90.35	98.14	92.75	89.05
Normalized Laplacian	97.01	97.88	95.12	89.30

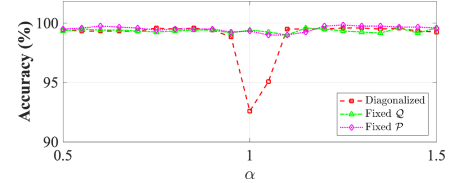


Fig. 11. Classification accuracy based on HGFrFT under different parameters $\alpha \in [0.5, 1.5]$.

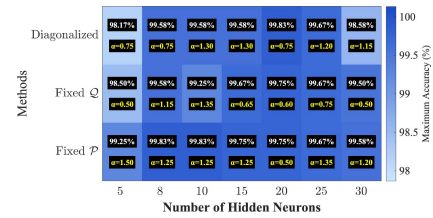


Fig. 12. Maximum classification accuracy based on HGFrFT under different parameters $\alpha \in [0.5, 1.5]$.

accuracy, showing that the fixed-operator variants can still extract useful fractional features. Fixed $\mathcal{Q}$ is stable across shift operators and remains close to the Diagonalized reference. Fixed $\mathcal{P}$ performs well for some operators but varies more across operators, which suggests greater sensitivity to the choice of shift operator in this setting.

Next, we evaluate the parameter sensitivity of three HGFrFT implementation methods: Diagonalized, Fixed $\mathcal{Q}$, and Fixed $\mathcal{P}$. Specifically, the fractional parameter α is varied from 0.5 to 1.5, and a neural network with 8 hidden neurons is used for classification. As shown in Fig. 11, each method exhibits a distinct optimal region with respect to α . Fixed $\mathcal{P}$ achieves the highest accuracy of 99.83% at $\alpha = 1.25$, while the Diagonalized and Fixed $\mathcal{Q}$ methods reach 99.58% at $\alpha = 0.75$ and 1.15, respectively. All methods consistently achieve accuracy above 99% across most parameter values, demonstrating robust performance and confirming the practical value of HGFrFT for ECG classification.

Furthermore, based on the simulation results in Fig. 12, we evaluate three HGFrFT variants across different numbers of hidden neurons. All methods achieve accuracies above 98%, demonstrating their robustness. The Diagonalized method reaches 99.83% accuracy with 20 neurons at $\alpha = 0.75$, Fixed $\mathcal{Q}$ achieves 99.75% with 20 neurons at $\alpha = 0.60$, and Fixed $\mathcal{P}$ attains 99.83% with 8 neurons at $\alpha = 1.25$. These results support the use of HGFrFT-based features for ECG classification and show that performance depends on the choice of α . Parameter selection should reflect both the application needs and the

available compute. In practice, we suggest searching $\alpha \in [0.5, 1.5]$ with a step size of 0.01 when higher accuracy is required. The final value can be chosen by cross-validation on the training set using the target performance metric.

VIII. CONCLUSION

This paper introduces the hyper-differential operator-based graph fractional Fourier transform (HGFrFT), providing a framework that bridges classical fractional Fourier transforms and graph signal processing. We derive the mathematical foundation of HGFrFT through the relationship between graph Fourier transform and fractional graph Fourier transform, establishing four distinct solutions to the Sylvester equation for constructing hyper-differential operators: Vectorized, Generic, Schur-based, and Diagonalized methods. Among these, Diagonalized and Schur-based provide accurate and efficient Sylvester-equation solvers for constructing the fractional transform operator, while the proposed fixed-operator forms offer lightweight alternatives when speed is prioritized over exact equivalence.

We also investigate parameter optimization for selecting the fractional order, providing a practical way to tune the transform for specific tasks. We further analyze the applications of HGFrFT in vertex-frequency analysis and optimal filter design. The proposed framework is validated through ECG signal classification experiments, where HGFrFT-based features demonstrate effective performance in distinguishing different heartbeat categories. The implementation code of this paper is publicly available at <https://github.com/chenjianyi888/HGFrFT>.

Future work will extend the window optimization framework to support task-driven objectives beyond mean squared error, such as classification loss and regularized criteria. This extension would make the proposed method applicable to a broader range of graph signal processing tasks.

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